

Live Queue & Token Realtime API – Backend Requirements

Overview:

A realtime queue management API should be implemented for handling live patient token flow using WebSocket.

The API should provide realtime updates for:

- Token Number
- Patient Name
- Estimated Consultation Time
- Current Running Token
- Queue Status

Important Requirement:

Entire system should work using WebSocket (Realtime).

No manual refresh or continuous polling should be required.

Recommended:

- Socket.IO
- Laravel WebSockets
- Pusher
- Firebase Realtime

Realtime Queue Update Logic:

The queue should automatically update when:

- Doctor starts consultation
- Doctor completes consultation
- Token gets skipped
- Appointment gets cancelled
- Token gets reassigned
- Queue timing shifts

Current Running Token:

System should maintain:

- Current consulting token
- Current patient details
- Session start time
- Estimated waiting queue

Auto Removal Logic:

Token should automatically be removed from active queue when:

1. Consultation Completed
2. Appointment Cancelled
3. Token Skipped/Reassigned

WebSocket Events Required:

- queue-updated
- doctor-session-started
- doctor-session-completed
- appointment-cancelled
- token-shifted

Recommended API:

GET /live-queue/{slot_id}

This API should return:

- current running token
- waiting queue
- estimated timings
- queue status

Queue Filtering Rules:

Only include:

- BOOKED
- CHECKED_IN
- WAITING
- IN_PROGRESS

Do not include:

- COMPLETED
- CANCELLED
- NO_SHOW

Important Backend Requirement:

This realtime queue system should work independently without affecting:

- appointment flow

- consultation flow
- billing flow
- receptionist workflow
- token generation logic

Recommended Architecture:

Backend:

- Central queue manager
- WebSocket broadcaster

Frontend:

- Realtime listener only
- No queue business logic handling

Summary:

The backend should provide:

- Realtime queue updates
- Automatic token removal
- Dynamic queue shifting
- WebSocket-based broadcasting
- Centralized queue management
- Minimal API usage with maximum realtime handling